

Complex Dynamics of Equal-Value Pole Maps

Anchored Divided Differences, Pole Geometry, and Anchor Bifurcations

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Abstract

For an analytic equation $F(z) = 0$, the anchored divided-difference map

$$P_{F,a}(z) = a - \frac{F(a)}{F[a, z]} = a - \frac{F(a)(a - z)}{F(a) - F(z)}$$

has roots of F as fixed points and has poles at the equal-value set $F(z) = F(a)$. In the complex plane this produces a meromorphic dynamical system whose pole geometry is fundamentally different from Newton dynamics. Newton poles are controlled by critical points of F ; anchored Pandrosion poles are controlled by level sets of F . This paper develops the basic complex dynamics of these equal-value pole maps. For polynomial F , $P_{F,a}$ is a rational map whose fixed points are exactly the roots of F away from the anchor, whose poles are the non-anchor points in $F^{-1}(F(a))$, and whose critical points solve

$$F(a) - F(z) - (a - z)F'(z) = 0.$$

The local multiplier at a root is

$$\lambda_r(a) = 1 - \frac{(a - r)F'(r)}{F(a)}.$$

Thus anchor motion changes both the local attraction at roots and the global pole skeleton. Figures for $F(z) = z^3 - 1$ show basin changes, equal-value pole motion, and anchor-parameter regions where all root multipliers are small. The result is a proposed new dynamical viewpoint: anchored maps encode the level-set geometry of F directly into their poles.

One-sentence summary. Anchored divided-difference dynamics replaces Newton's critical-point pole geometry by equal-value pole geometry: the poles are $F^{-1}(F(a))$.

1 Introduction

The scalar anchored Pandrosion map

$$P_{F,a}(z) = a - \frac{F(a)}{F[a, z]}$$

is a fixed-anchor secant map. On the real line its main role is local residual control. Over the complex plane it becomes a meromorphic map, and the location of its poles becomes a central dynamical object.

The essential difference from Newton's method is immediate. Newton's map is

$$N_F(z) = z - \frac{F(z)}{F'(z)},$$

so its poles occur where $F'(z) = 0$. The anchored map is instead

$$P_{F,a}(z) = a - \frac{F(a)(a-z)}{F(a) - F(z)},$$

so its poles occur where

$$F(z) = F(a).$$

Thus the anchor does not merely set a local contraction factor; it selects a level set of F and inserts that level set into the global rational dynamics.

Main questions. This paper organizes the first layer of this dynamics:

- (i) Which fixed points and poles does $P_{F,a}$ have?
- (ii) How does the root multiplier change with the anchor?
- (iii) What are the critical points of the rational map?
- (iv) How do basins change as the anchor moves?

2 The equal-value pole map

Let F be a nonconstant polynomial or meromorphic function and choose an anchor a with $F(a) \neq 0$.

Definition 2.1 (Equal-value pole map). The anchored equal-value pole map is

$$P_{F,a}(z) = a - \frac{F(a)(a-z)}{F(a) - F(z)}. \tag{1}$$

When $z = a$ is a removable point and $F'(a) \neq 0$, one inserts the value

$$P_{F,a}(a) = a - \frac{F(a)}{F'(a)}.$$

Proposition 2.2 (Fixed points and poles). *Assume $F(a) \neq 0$.*

- (a) *If $F(r) = 0$ and $r \neq a$, then $P_{F,a}(r) = r$.*
- (b) *If $P_{F,a}(z) = z$ and $z \neq a$ is not a pole, then $F(z) = 0$.*
- (c) *The non-removable poles are contained in*

$$\Pi_{F,a} = \{z \neq a : F(z) = F(a)\}.$$

Proof. The fixed-point proof is the scalar divided-difference proof. If $F(r) = 0$, then $F[a, r] = F(a)/(a-r)$, so $P_{F,a}(r) = r$. Conversely, $P_{F,a}(z) = z$ gives $F(a) - F(z) = F(a)$, hence $F(z) = 0$. The pole statement follows directly from the denominator in (1). $\square$

Remark 2.3. For $F(z) = z^p - x$ and anchor $a = 1$, the equal-value poles satisfy $z^p = 1$. The non-anchor poles are therefore the nontrivial roots of unity, recovering the cyclotomic pole geometry of the monomial theory.

3 Rational form, multipliers, and critical points

If F is a polynomial, then

$$P_{F,a}(z) = \frac{F(a)z - aF(z)}{F(a) - F(z)}. \quad (2)$$

This is a rational map on $\hat{C}$ after removable factors are cancelled.

Proposition 3.1 (Root multiplier). *If r is a simple root of F and $a \neq r$, then*

$$P'_{F,a}(r) = 1 - \frac{(a-r)F'(r)}{F(a)}.$$

Proof. Differentiate (1) or use the exact anchored multiplier identity. $\square$

Proposition 3.2 (Critical equation). *Away from poles and removable points, the critical points of $P_{F,a}$ satisfy*

$$F(a) - F(z) - (a-z)F'(z) = 0. \quad (3)$$

Proof. Let $H(z) = F(a) - F(z)$. From (1),

$$P'_{F,a}(z) = F(a) \frac{H(z) - (a-z)F'(z)}{H(z)^2}.$$

Thus critical points satisfy (3). $\square$

Remark 3.3. Equation (3) says that the critical points are those where the secant slope from the anchor equals the tangent slope at z :

$$\frac{F(a) - F(z)}{a - z} = F'(z).$$

Thus criticality is also a geometric secant-tangent condition.

4 Anchor bifurcation

The anchor parameter a affects the dynamics in two coupled ways:

- (1) it changes the root multipliers $\lambda_r(a)$;
- (2) it moves the equal-value pole set $\Pi_{F,a}$.

This is unlike Newton dynamics, where the pole set is tied to F' and does not move with an external anchor.

Definition 4.1 (Multiplier profile). For a polynomial with simple roots $r_1, \dots, r_d$, define

$$\Lambda(a) = \max_{1 \leq j \leq d} \left| 1 - \frac{(a-r_j)F'(r_j)}{F(a)} \right|.$$

A root-uniformly attracting anchor is one with $\Lambda(a) < 1$.

This definition is local. It does not guarantee global convergence, because global basins can be interrupted by equal-value poles. It does identify anchor regions where all roots are locally attracting.

5 Example: $F(z) = z^3 - 1$

Let $F(z) = z^3 - 1$. The roots are $1, \zeta, \zeta^2$, where $\zeta = e^{2\pi i/3}$. For an anchor a with $a^3 \neq 1$, the non-removable equal-value poles solve

$$z^3 = a^3, \quad z \neq a,$$

so they are

$$\zeta a, \quad \zeta^2 a.$$

Thus pole motion is completely visible: moving the anchor rotates two pole companions around it.

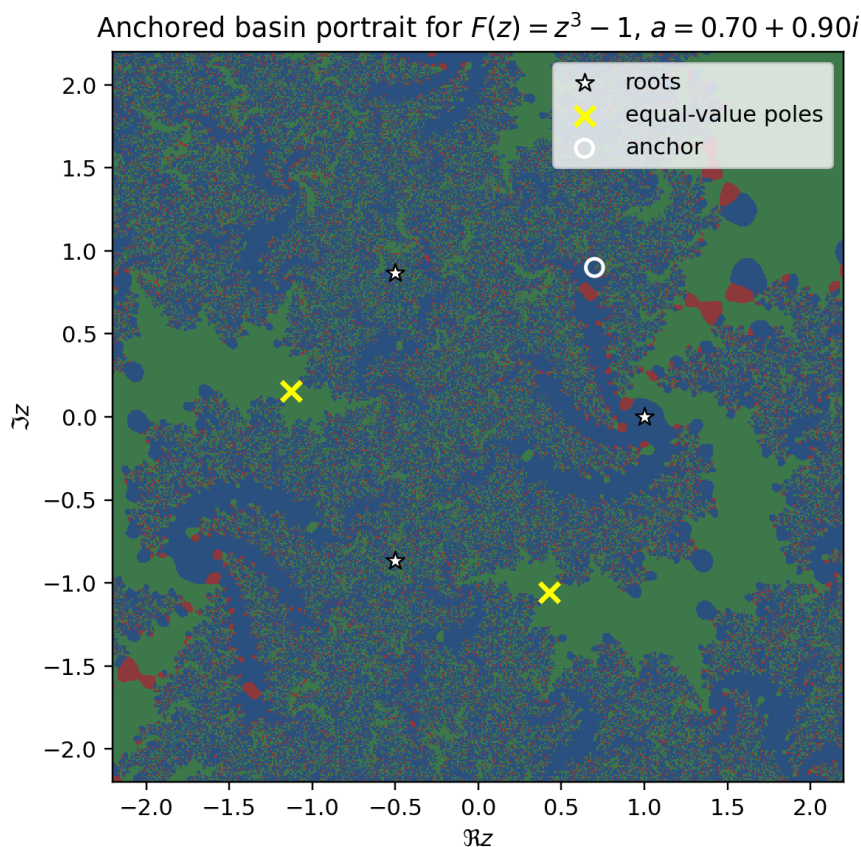


Figure 1: Basins for $F(z) = z^3 - 1$ with anchor $a = 0.70 + 0.90i$. The star markers are roots; yellow crosses mark the equal-value poles ζa and $\zeta^2 a$.

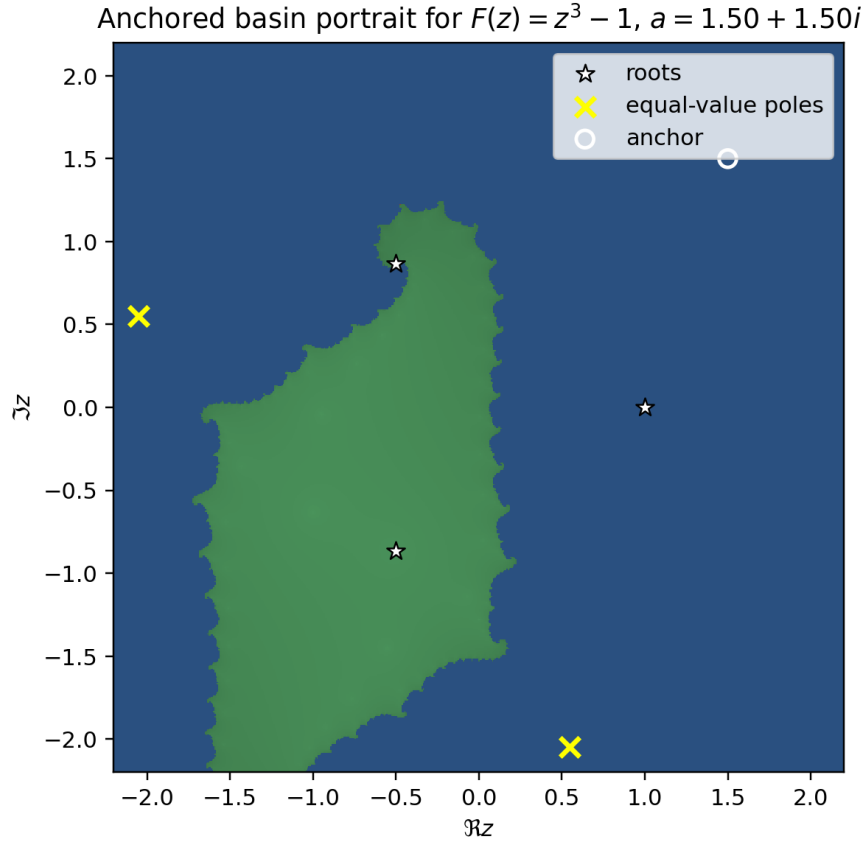


Figure 2: Changing the anchor changes both the pole skeleton and the basin geometry. The poles are not fixed critical points of F ; they are points on the anchor level set $F^{-1}(F(a))$.

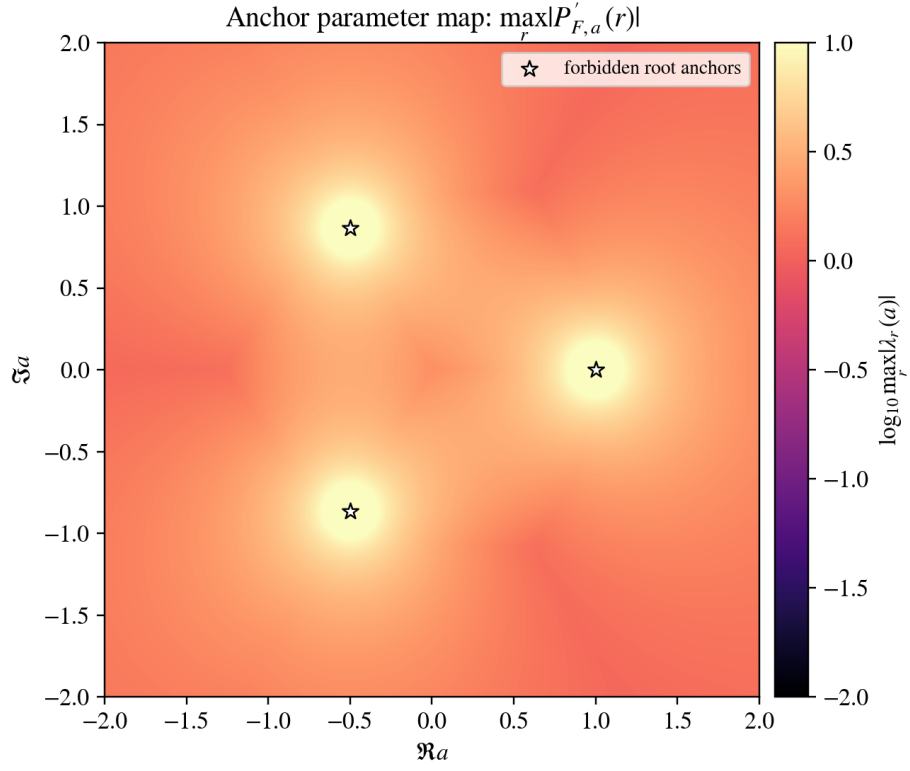


Figure 3: Anchor-parameter map for $\Lambda(a) = \max_r |\lambda_r(a)|$. Darker regions have smaller worst root multiplier. The roots are excluded anchor values because $F(a) = 0$.

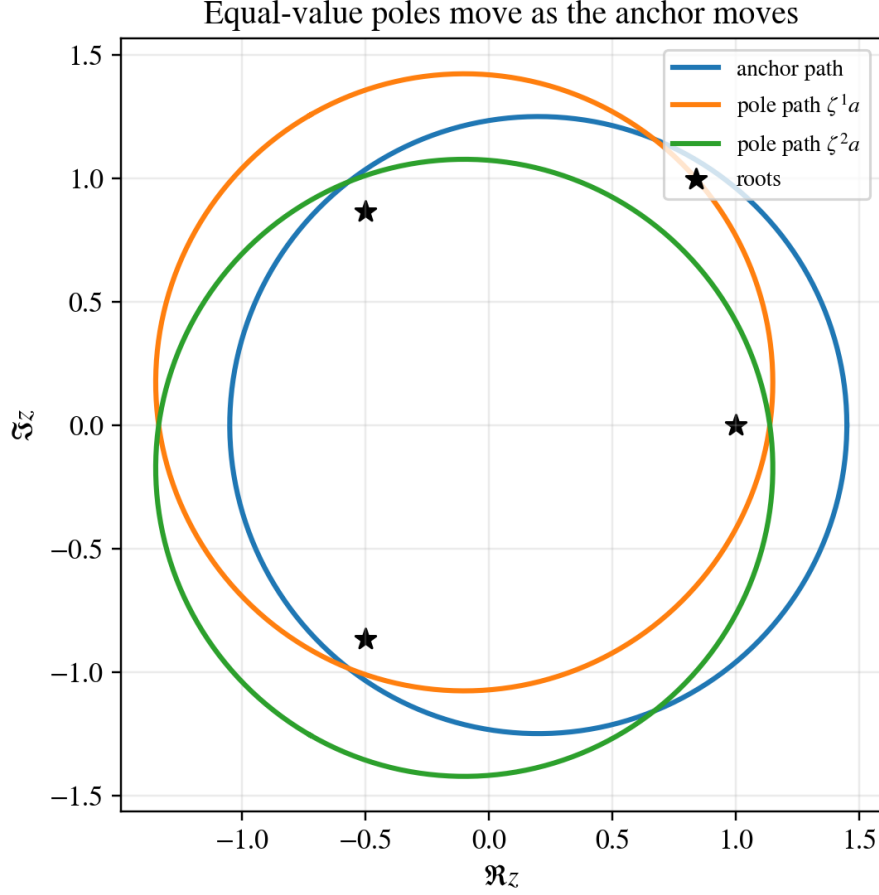


Figure 4: Equal-value pole motion for $F(z) = z^3 - 1$. Along an anchor path, the two pole companions are ζa and $\zeta^2 a$. This is the simplest visible form of equal-value pole geometry.

6 Comparison with Newton dynamics

Newton's method and anchored divided-difference dynamics share roots as fixed points, but their global skeletons are different:

$$N_F(z) = z - \frac{F(z)}{F'(z)} \Rightarrow \text{poles from } F'(z) = 0,$$

whereas

$$P_{F,a}(z) = a - \frac{F(a)(a - z)}{F(a) - F(z)} \Rightarrow \text{poles from } F(z) = F(a).$$

Newton's pole geometry is intrinsic to the critical set of F . Anchored pole geometry is extrinsic: it depends on the selected value $F(a)$.

This difference suggests that anchored dynamics may be useful not as a faster root finder, but as a dynamical probe of level sets. The map visualizes the interaction between roots, equal-value sets, and secant-tangent criticality.

7 Open problems

The equal-value viewpoint suggests several questions.

- (i) Classify the Fatou components created by moving equal-value poles.

- (ii) Determine when anchor motion causes pole-root collisions or basin boundary bifurcations.
- (iii) Study the measure of anchors for which all roots are locally attracting.
- (iv) Extend the pole-set description to meromorphic functions and maps on Riemann surfaces.
- (v) Compare the critical equation (3) with the ramification of the level-set covering $F : \widehat{C} \rightarrow \widehat{C}$.

8 Conclusion

Anchored divided-difference maps have a distinctive complex dynamics: their poles are equal-value points. This makes the anchor a global dynamical parameter, not only a local convergence parameter. The resulting maps encode the level-set geometry of F directly into the rational dynamics. This provides a new and natural continuation of the Pandrosion program into holomorphic dynamics.

References

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